Outline: the massless phase of the $\mathbb{H}^{0|4}$ model in $d = 5$

(adaptation of `h02rg`; draft)

1 November 2026

Scope. We outline a proof that the $\mathbb{H}^{0|4}$ model on $\mathbb{Z}^d$, $d \geq 5$ (target: $d = 5$), has spontaneous symmetry breaking with massless free-field low temperature fluctuations for β large, by adapting `h02rg/src/h02rg.tex` (arXiv:2107.01878; cited as [h02rg] with its LaTeX labels). Only the order of limits $\lim_{h\downarrow 0} \lim_{N\to\infty}$ is treated (the analogue of [h02rg, `thm:h02`]); the finite volume order of limits ([h02rg, `thm:psi4-full`]) is deferred.

Status tags. [V]: verbatim from [h02rg] (up to the number of Grassmann generators per site); [M]: minor change, indicated; [N]: new. [done]: argument written out here and needs only checking (in particular of sign conventions).

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1 The model and the target theorem

For each $x \in \Lambda$ let ξ_x^a, η_x^a $(a = 1, 2)$ be generators of a Grassmann algebra and set

$$s_x = \sum_{a=1}^{2} \xi_x^a \eta_x^a, \qquad q_x = \xi_x^1 \eta_x^1 \xi_x^2 \eta_x^2, \qquad z_x = \sqrt{1 - 2s_x}, \qquad u_x = (\xi_x^1, \eta_x^1, \xi_x^2, \eta_x^2, z_x), \tag{1}$$

$$u_x \cdot u_y = -\sum_{a=1}^{2} (\xi_x^a \eta_y^a + \xi_y^a \eta_x^a) - z_x z_y. \tag{2}$$

With $\frac{1}{2}(u, \Delta u) = \sum_{xy \in E(\Lambda)} (u_x \cdot u_y + 1)$ as in [h02rg, Section 1.2], define

$$\langle F \rangle_{\beta,h} = \frac{1}{Z_{\beta,h}} \int \Big(\prod_{x \in \Lambda} \partial_{\eta_x^2} \partial_{\xi_x^2} \partial_{\eta_x^1} \partial_{\xi_x^1} \frac{1}{z_x} \Big) e^{\frac{\beta}{2}(u, \Delta u) - h(1, z-1)} F. \tag{3}$$

Remark 1.1. (i) Unlike $\mathbb{H}^{0|2}$ there is no known (to us) probabilistic representation, so $Z_{\beta,h} \neq 0$ is not automatic; it follows from the renormalisation group (RG) representation in the regime considered. (ii) The symmetry group is $OSp(1|4)$; its even part $Sp(4)$ acts on (ξ^a, η^a) and preserves s, hence z.

Theorem 1.2 (Target)**.** *Let $d \geq 5$ and $L \geq L_0(d)$. There are $\beta_0 < \infty$, $\kappa > 0$, $\theta_d(\beta) = 1 + O(1/\beta)$ and $c_i(\beta) = c_i + O(1/\beta)$ such that for $\beta \geq \beta_0$, $a, b \in \{1, 2\}$,*

$$\lim_{h \downarrow 0} \lim_{N \to \infty} \langle z_0 \rangle_{\beta,h} = \theta_d(\beta), \tag{4}$$

$$\lim_{h \downarrow 0} \lim_{N \to \infty} \langle \xi_0^a \eta_x^b \rangle_{\beta,h} = \delta_{ab} \frac{c_1(\beta)}{\beta |x|^{d-2}} + O\Big(\frac{1}{\beta |x|^{d-2+\kappa}} \Big), \tag{5}$$

$$\lim_{h \downarrow 0} \lim_{N \to \infty} \langle z_0; z_x \rangle_{\beta,h} = -\frac{c_2(\beta)}{\beta^2 |x|^{2d-4}} + O\Big(\frac{1}{\beta^2 |x|^{2d-4+\kappa}} \Big), \tag{6}$$

with $c_1 = c_d$ and (to be confirmed) $c_2 = 2c_d^2$, the factor 2 being the number of flavours.

2 Fermionic reformulation [N, done]

Lemma 2.1. *For each x: $s^2 = 2q$, $s^3 = sq = 0$; $z = 1 - s - q$; $u \cdot u = -1$; $1/z = e^{s+2q}$; and*

$$u_x \cdot u_y + 1 = \sum_a (\xi_x^a - \xi_y^a)(\eta_x^a - \eta_y^a) + \tfrac{1}{2}(\nabla s)^2 + (\nabla s)(\nabla q) + \tfrac{1}{2}(\nabla q)^2, \tag{7}$$

where $\nabla f = f_x - f_y$.

Proof. The $\xi^a \eta^a$ are even and square to zero, so $s^2 = \xi^1 \eta^1 \xi^2 \eta^2 + \xi^2 \eta^2 \xi^1 \eta^1 = 2q$, and every product of s with q repeats a generator, so $sq = 0$ and $s^3 = 2sq = 0$. Hence $\sqrt{1 - 2s} = 1 - s - \frac{1}{2}s^2 = 1 - s - q$; check: $(1 - s - q)^2 = 1 - 2s - 2q + s^2 = 1 - 2s$. Then $u \cdot u = -2s - z^2 = -1$. Next, $-\frac{1}{2}\log(1 - 2s) = s + s^2 = s + 2q$, so $1/z = (1 - 2s)^{-1/2} = e^{s+2q}$. Finally, by bilinearity and $u_x \cdot u_x = u_y \cdot u_y = -1$, $u_x \cdot u_y + 1 = -\frac{1}{2}(u_x - u_y) \cdot (u_x - u_y) = \sum_a \nabla \xi^a \nabla \eta^a + \frac{1}{2}(\nabla z)^2$, and $\nabla z = -\nabla s - \nabla q$ gives (7). □

Corollary 2.2. *The density in (3) equals*

$$\exp\Big[\beta \sum_{xy} \Big(\sum_a \nabla \xi^a \nabla \eta^a + \tfrac{1}{2}(\nabla s)^2 + (\nabla s)(\nabla q) + \tfrac{1}{2}(\nabla q)^2 \Big) + \sum_x \big((1 + h) s_x + (2 + h) q_x \big) \Big], \tag{8}$$

since $-h(z - 1) = h(s + q)$ and $1/z = e^{s+2q}$. (For $q \equiv 0$ this is [h02rg, `e:h02-psi`] *with the familiar $(1 + h)$.)*

Rescaling. For $s_0 > -1$ put $\varepsilon = (1+s_0)/\beta$, $\xi^a = \sqrt{\varepsilon}\,\bar\psi^a$, $\eta^a = \sqrt{\varepsilon}\,\psi^a$, and

$$S = \sum_a \psi^a\bar\psi^a, \qquad Q = \psi^1\bar\psi^1\psi^2\bar\psi^2 = \tfrac12 S^2, \qquad s = -\varepsilon S, \quad q = \varepsilon^2 Q, \quad z = 1+\varepsilon S - \varepsilon^2 Q. \tag{9}$$

With $b_0 = \beta\varepsilon^2 = (1+s_0)\varepsilon$ the exponent of Corollary 2.2 becomes

$$-(1+s_0)\sum_{xy}\sum_a \nabla\psi^a\nabla\bar\psi^a - (1+h)\varepsilon\sum_x S_x + (2+h)\varepsilon^2\sum_x Q_x + \frac{b_0}{2}\sum_{xy}(\nabla S)^2 - b_0\varepsilon\sum_{xy}(\nabla S)(\nabla Q) + \frac{b_0\varepsilon^2}{2}\sum_{xy}(\nabla Q)^2. \tag{10}$$

This is of the form $e^{-\sum_a(\psi^a,(-\Delta+m^2)\bar\psi^a)-V_0(\Lambda)-W_0(\Lambda)}$ (sign conventions for the Gaussian part to be matched with [h02rg, `e:psi4`]), with

$$V_0 = \sum_x\Big[s_0\sum_a(\nabla\psi^a)_x(\nabla\bar\psi^a)_x + a_0 S_x + g_0 Q_x + b_0 B_x\Big], \qquad B_x = -\tfrac14\sum_{e\in\mathcal{E}_d}(\nabla_e S)_x^2, \tag{11}$$

$$a_0 = (1+h)\varepsilon - m^2, \qquad g_0 = -(2+h)\varepsilon^2, \qquad b_0 = (1+s_0)\varepsilon, \tag{12}$$

$$W_0 = \sum_{xy}\Big[b_0\varepsilon(\nabla S)(\nabla Q) - \tfrac{b_0\varepsilon^2}{2}(\nabla Q)^2\Big]. \tag{13}$$

Remark 2.3 (Choice of the b-monomial). For $\mathbb{H}^{0|2}$ the b-monomial of [h02rg, `e:quartic`] is $\psi_x\bar\psi_x\cdot\frac12\sum_e\psi_{x+e}\bar\psi_{x+e}$, which coincides with a squared gradient because $S^2 = 0$. For $\mathbb{H}^{0|4}$, $S_xS_{x+e} = 2Q_x + S_x\nabla_e S_x$ contains the local monomial Q, so the b-monomial must be the squared gradient B in (11). Harmless in $d \geq 5$, essential in $d \leq 4$.

3 Ward identity [N, done]

Lemma 3.1. *Let $T_a = \sum_x z_x\partial_{\xi_x^a}$ (an odd derivation). Then $T_a z_x = -\eta_x^a$, $T_a(u_x\cdot u_y) = 0$, and $\int d\mu\, T_a G = 0$ for every G, where $d\mu = \prod_x \partial_{\eta_x^2}\partial_{\xi_x^2}\partial_{\eta_x^1}\partial_{\xi_x^1} z_x^{-1}$. Consequently, for F of parity $|F|$,*

$$\langle T_a F\rangle_{\beta,h} = -(-1)^{|F|}\,h\sum_{x\in\Lambda}\langle F\eta_x^a\rangle_{\beta,h}. \tag{14}$$

In particular, with $F = \xi_0^a$,

$$\langle z_0\rangle_{\beta,h} = h\sum_{x\in\Lambda}\langle\xi_0^a\eta_x^a\rangle_{\beta,h} \qquad (\textit{no sum over } a). \tag{15}$$

Proof. Since z_x is a function of s_x and $\partial_{\xi_x^a}s_x = \eta_x^a$, we get $\partial_{\xi_x^a}z_x = -z_x^{-1}\eta_x^a$, so $T_a z_x = -\eta_x^a$. Then $T_a(\xi_x^a\eta_y^a + \xi_y^a\eta_x^a) = z_x\eta_y^a + z_y\eta_x^a$ and $T_a(z_xz_y) = -\eta_x^a z_y - z_x\eta_y^a$, which cancel in $u_x\cdot u_y$. For the integral: $\big(\prod_y z_y^{-1}\big)z_x\partial_{\xi_x^a}G = \partial_{\xi_x^a}\big(\prod_{y\neq x}z_y^{-1}G\big)$ because $\prod_{y\neq x}z_y^{-1}$ is even and independent of ξ_x^a; the Berezin integral of a derivative vanishes. Apply this to $G = Fe^A$ with $A = \frac{\beta}{2}(u,\Delta u) - h(1,z-1)$: $T_aA = h\sum_x\eta_x^a$ and $T_a(Fe^A) = (T_aF)e^A + (-1)^{|F|}F(T_aA)e^A$. □

Remark 3.2 (Second identity, not needed for $d \geq 5$). $F = \xi_0^1\xi_0^2\eta_0^2$ gives $T_1F = z_0\xi_0^2\eta_0^2 = \xi_0^2\eta_0^2 - q_0$, hence $\langle\xi_0^2\eta_0^2 - q_0\rangle_{\beta,h} = h\sum_x\langle\xi_0^1\xi_0^2\eta_0^2\eta_x^1\rangle_{\beta,h}$. This is the candidate for pinning the Q-coupling in $d \leq 4$.

4 Power counting, symmetry and localisation [M]

Symmetry. Replace $U(1)$ invariance of [h02rg, `sec:symmetries`] by invariance under $G_{\mathrm{fl}} = U(1)^2 \rtimes S_2$ (separate phase rotations of the two flavours and the flavour swap), a subgroup of $Sp(4)$. $\mathbb{E}_C\theta$ preserves G_{fl}-invariance because C is flavour-diagonal. The bare potential is $Sp(4)$-invariant.

Power counting. $[\psi] = (d-2)/2$; a monomial with p fields and k gradients has dimension $p(d-2)/2 + k$. Odd k is excluded by lattice symmetries, as in [h02rg].

monomial	dimension	$d = 5$	status in $\mathcal{V}$
S (a)	$d-2$	3	relevant
$\sum_a \nabla\psi^a\nabla\bar\psi^a$, $\frac{1}{2}\sum_a(\psi^a(-\Delta\bar\psi^a) + (-\Delta\psi^a)\bar\psi^a)$ (y,z)	d	5	marginal
Q (g)	$2(d-2)$	6	irrelevant, carried
B (b)	$2(d-2)+2$	8	irrelevant, carried
$(\nabla S)(\nabla Q)$, $(\nabla Q)^2$	$3(d-2)+2$, $4(d-2)+2$	11, 14	irrelevant, in K_0

Definition 4.1. $\mathcal{V}^{\varnothing} \cong \mathbb{C}^5$: formal local polynomials $V = y\sum_a \nabla\psi^a\nabla\bar\psi^a + \frac{z}{2}\sum_a(\psi^a(-\Delta\bar\psi^a) + (-\Delta\psi^a)\bar\psi^a) + aS + gQ + bB$.

Proposition 4.2 (analogue of [h02rg, `prop:Loc`])**.** *With field dimensions as above and $d_+ = d$, the operators* $\mathrm{Loc}_X, \mathrm{Loc}_{X,Y}$ *of [h02rg] (from [MR3332939]) have the same properties, with image in the (z, y, a) part of $\mathcal{V}^{\varnothing}$. In $d \geq 5$ every monomial of degree ≥ 4 is irrelevant, so* Loc *never produces b or g.*

Lemma 4.3 (Wick computations)**.** *With the convention $\mathbb{E}_C[\bar\zeta_x\zeta_y] = C(x,y)$, so $\mathbb{E}_C\theta(\psi^a\bar\psi^a) = \psi^a\bar\psi^a - C(0)$, and since C is flavour-diagonal,*

$$\mathbb{E}_C\theta Q = Q - C(0)S + C(0)^2, \qquad \mathbb{E}_C\theta B = B + (\textit{quadratic terms as in } [\text{h02rg, } \texttt{eq:tildeflow}], \textit{ summed over } a) + \textit{con} \tag{16}$$

The second identity uses that $(\nabla_e S)^2 = \sum_{a,b}\nabla_e P^a \nabla_e P^b$ with $P^a = \psi^a\bar\psi^a$, and that for $a \neq b$ the factors are independent under $\mathbb{E}_C$, while $\mathbb{E}_C\theta\nabla_e P^a = \nabla_e P^a$. Hence the flow before Loc *is*

$$\tilde z = z + \kappa^{zb} b, \quad \tilde y = y + \kappa^{yb} b, \quad \tilde a = a + \kappa^{ab} b - C_{j+1}(0)\, g, \quad \tilde b = b, \quad \tilde g = g, \tag{17}$$

with the κ's of [h02rg, `eq:tildeflow`] *(to be rechecked: the flavour sum may change κ^{ab} by a factor).*

Remark 4.4 (Why W_0 goes into K_0). Single contractions of $(\nabla S)(\nabla Q)$ produce degree-4 monomials outside $\mathrm{span}\{Q, B\}$, so the W_0-monomials do not span an $\mathbb{E}_C\theta$-invariant space. Placing W_0 into the initial polymer activity K_0 avoids enlarging $\mathcal{V}^{\varnothing}$.

5 Bulk renormalisation group flow [M]

(a) Finite range decomposition [h02rg, `sec:frd`]: [V], applied flavour-diagonally.

(b) Grassmann Gaussian integration and polymer coordinates: [V].

(c) Norms [h02rg, `sec:norms`]: [M]. Same T_ϕ norms; constants depend on the number of generators per site (4 instead of 2). Block norms: $\|V_j(B)\| \asymp |z_j| + |y_j| + L^{2j}|a_j| + L^{-(d-2)j}|b_j| + L^{-(d-4)j}|g_j|$.

(d) RG map [h02rg, `sec:step-def`], including [h02rg, `prop:ZVK`] and [h02rg, `prop:consistency`]: [V].

(e) Step estimates [h02rg, `thm:step`]: [M]. The new input is g; check every place where the explicit form of V enters (in particular the stability estimate for e^{-V} and the second-order Loc terms).

(f) **Initial activity** [N, routine]: with $f_{xy} = e^{-w_{xy}} - 1$, where $W_0 = \sum_{xy} w_{xy}$,

$$e^{-V_0(\Lambda)-W_0(\Lambda)} = \sum_{X\subset\Lambda} e^{-V_0(\Lambda\setminus X)} K_0(X), \qquad K_0(X) = e^{-V_0(X)} \sum_{E'\subset E:\ V(E')=X} \prod_{xy\in E'} f_{xy}. \quad (18)$$

K_0 factorises over connected components, is G_{fl}- and lattice-invariant, and $\|K_0\|_0 = O(b_0\varepsilon)$. To check: that K_0 lies in the scale-0 space $\mathcal{K}_0$ of [h02rg] (regulators at scale 0).

Theorem 5.1 (analogue of [h02rg, `thm:flow`])**.** *Let $d \geq 5$, $L \geq L_0$, $A \geq A_0(L)$. For $m^2 \geq 0$ and $w_0 = (b_0, g_0, K_0)$ with $|b_0| + |g_0| + \|K_0\|_0 \leq \delta$, there is $v_0^c(m^2, w_0) = (z_0, y_0, a_0)$ such that the flow from (v_0^c, w_0) satisfies $\|V_j\|_j = O_L(\delta L^{-\kappa j})$ and $\|K_j\|_j = O_L(\delta L^{-\kappa j})$. v_0^c is continuous and bounded in $m^2 \geq 0$, and analytic in w_0 with uniformly bounded derivative.*

Proof sketch. As in [h02rg], with $v_j = (\hat{z}_j, \hat{y}_j, \hat{a}_j)$ and $w_j = (\hat{b}_j, \hat{g}_j, K_j)$, where $\hat{g}_j = L^{-(d-4)j} g_j$. Then $\hat{g}_{j+1} = L^{-(d-4)}\hat{g}_j$ joins D_j, and $\|D_j\| \leq \max(L^{-(d-2)}, L^{-(d-4)}, O(L^{-3} + A^{-\eta}))$, which is < 1 for $d \geq 5$. The new entry of B_j is $\hat{\kappa}_j^{ag}\hat{g}_j$ with $\hat{\kappa}_j^{ag} = -L^{2(j+1)}L^{(d-4)j}C_{j+1}(0) = O(L^2)$ by [h02rg, `e:C-bd`]. Apply the stable manifold theorem (Lean: `Flow.System.Bounds.existsUnique_stable`, `analyticAlongCurves_stableMap`, `continuous_stableMap`). Note w_0 now has a nonzero K_0 component, which the abstract theorem allows. □

Corollary [h02rg, `cor:flow-finvol`] and susceptibility ([h02rg, `prop:ZNN`], [h02rg, `prop:suscept`]). [M]: per flavour, $\sum_x \langle \bar{\psi}_0^a \psi_x^a \rangle = m^{-2} + O(\delta L^{-(2+\kappa)N}) m^{-4}$.

6 Observables [M]

- *Case (1)*, $\langle \bar{\psi}_a^1 \psi_b^1 \rangle$: [V] with flavour-1 sources $\sigma_a \bar{\psi}_a^1$, $\psi_b^1 \bar{\sigma}_b$. Since $d_+^a = (d-2)/2$, the cubic terms $\sigma_a \bar{\psi}_a^1 P_a^2$ (dimension $3(d-2)/2$) are irrelevant and not extracted.

- *Case (2)*: sources $\sigma_x = \bar{\vartheta}_x \vartheta_x$ at $x \in \{a, b\}$ coupled to $f_x \in \mathrm{span}\{S_x, Q_x\} = \mathrm{span}\{z_x - 1, (z_x - 1)^2\}$, i.e. to the $Sp(4)$-invariant local functions. Loc^a with $d_+^a = d - 2$ extracts $\sigma_a(\nu + \lambda S_a)$; $\sigma_a Q_a$ (dimension $2(d-2)$) is irrelevant and is placed in the initial observable activity ([N, routine], as for K_0).

- Result (analogue of [h02rg, `prop:1pt`], [h02rg, `prop:2pt`], $N \to \infty$ at $m^2 > 0$, then continuity in $m^2 \geq 0$): for $f \in \{S, Q\}$ there are γ_f, λ_f continuous in (w_0, m^2) with

$$\langle f_0 \rangle = \gamma_f, \qquad \langle f_0; g_x \rangle = -2\lambda_f \lambda_g \, (-\Delta + m^2)^{-1}(0, x)^2 + O(\delta |x|^{-2(d-2)-\kappa}), \quad (19)$$

with $\lambda_S = 1 + O(\delta)$, and λ_Q, γ_Q given at leading order by the free Wick computation. The factor 2 is the flavour sum for S; it is to be confirmed together with the signs.

7 Matching with $\mathbb{H}^{0|4}$ via the Ward identity [M]

Let $s_0^c(w_0, m^2)$ and $a_0^c(w_0, m^2)$ be the critical couplings of Theorem 5.1 (with $s_0 = y_0 + z_0$ as in [h02rg]). By (12)–(13), $\mathbb{H}^{0|4}$ corresponds to $w_0 = w_0(\varepsilon, h, s_0)$ with $b_0 = (1 + s_0)\varepsilon$, $g_0 = -(2 + h)\varepsilon^2$ and $K_0 = K_0[W_0]$ with coefficients $O(\varepsilon^2)$.

Step 1. (s_0). For fixed (ε, h, m^2) solve $s_0 = s_0^c(w_0(\varepsilon, h, s_0), m^2)$. The right side is Lipschitz in s_0 with constant $O(\varepsilon)$, so Banach's fixed point theorem applies.

Step 2. (h). Solve $(1+h)\varepsilon = a_0^c(w_0(\varepsilon, h, s_0(h)), m^2) + m^2$ for h. The map $h \mapsto \varepsilon^{-1}(a_0^c + m^2) - 1$ is Lipschitz with constant $O(\varepsilon)$, since h enters only through $g_0 = -(2+h)\varepsilon^2$ and through s_0; restrict to $m^2 \le m^2_{\max}$. This defines $h(\varepsilon, m^2)$ and $\beta(\varepsilon, m^2) = (1 + s_0)/\varepsilon$. (New compared with [h02rg, `e:hb0m`]: there h is explicit; here the irrelevant input depends on h.)

Step 3. (Ward). By (15) and Section 6 in the limit $N \to \infty$ with $m^2 > 0$: $\langle z_0 \rangle = 1 + \varepsilon\gamma_S - \varepsilon^2\gamma_Q =: \Theta(\varepsilon, m^2)$ and $h \sum_x \langle \xi_0^1 \eta_x^1 \rangle = h\varepsilon\, m^{-2}$. Hence

$$h(\varepsilon, m^2) = \frac{m^2\, \Theta(\varepsilon, m^2)}{\varepsilon} = m^2 \beta(\varepsilon, m^2)\,(1 + O(\varepsilon)), \tag{20}$$

so $h(\varepsilon, 0) = 0$ and $h > 0$ for $m^2 > 0$. This is [h02rg, `cor:ward`](i).

Step 4. (Inversion). As [h02rg, `cor:ward`](ii) / [MR3339164, Prop. 4.2]: for β large and $h \ge 0$ there are $\tilde{\varepsilon}(\beta, h), \tilde{m}^2(\beta, h)$, right-continuous as $h \downarrow 0$. This requires differentiability in ε (from analyticity in w_0) and continuity in m^2.

8 Proof of Theorem 1.2 [M]

Since $\xi_0^a \eta_x^b = \varepsilon \bar{\psi}_0^a \psi_x^b$ and $z = 1 + \varepsilon S - \varepsilon^2 Q$, Sections 5–7 give, after $N \to \infty$ and then $h \downarrow 0$ (equivalently $m^2 \downarrow 0$):

- $\theta_d(\beta) = \Theta(\varepsilon, 0) = 1 + O(\varepsilon)$;
- $c_1(\beta) = (1 + s_0^c)c_d$, using $\langle \bar{\psi}_0^a \psi_x^b \rangle = \delta_{ab} G(x) + \dots$ (flavour-diagonality from G_{fl});
- $\langle z_0; z_x \rangle = -2(\varepsilon\lambda_S - \varepsilon^2\lambda_Q)^2 G(x)^2 + \dots$, so $c_2(\beta) = 2(1 + s_0^c)^2(\lambda_S - \varepsilon\lambda_Q)^2 c_d^2$.

9 Deferred

- Finite volume order of limits ([h02rg, `thm:psi4-full`], [h02rg, `sec:finvol`]): zero mode integration now involves four Grassmann zero modes.
- Analogue of Appendix A (high temperature, $h > 0$): no forest representation is available.
- $d = 4$: $\hat{g}_{j+1} = \hat{g}_j + O(\hat{g}^2, \hat{g}\hat{b}, \dots)$ is marginal, so $\|D_j\| < 1$ fails in the g-direction. This is where the second Ward identity enters.

10 Lean plan

`Grassmann/*`	Gap G1 in `docs/duplication-audit.md`: generator index $\Lambda \times \iota \times \mathbb{Z}^\times$ with $\iota = \mathrm{Fin}\, 2$.
`Loc/Stencil`, `Loc2`, `Loc4`	Gap G2: new coupling structure (z, y, a, b, g) and evaluation; Loc unchanged in $d \geq 5$ beyond monomial definitions.
`H02RG/Step/*`	Wick computation (Lemma 4.3); norm constants; stability with g.
`H02RG/Flow/*`	`StableManifold.lean` reused as is; new `Theorem.lean` with $w = (\hat{b}, \hat{g}, K)$ and $K_0 \neq 0$; initial-activity lemma.
`H02RG/Obs/*`, `PtCorr`	Gap G5: Case (2) observables $\sigma_a f_a$ with $f \in \{S, Q\}$.
new `H04/Ward.lean`	Lemma 3.1; the $\mathbb{H}^{0\|2}$ version is in `H02/Ward.lean`.
new `H04/Main`	`thm_h04`, the analogue of `H02RG.thm_h02`, with hypothesis `5 <= d`.

11 Checks before writing proofs

1. Sign and ordering conventions in Corollary 2.2, (10) and (14) against [h02rg] and `H02/Ward.lean`.
2. κ-coefficients with the flavour sum (Lemma 4.3).
3. Initial activity $K_0 \in \mathcal{K}_0$ and the observable analogue; whether any step of [h02rg, `thm:step`] uses $K_0 = 0$ beyond the initial condition.
4. The flavour factor in c_2.
5. Signs of γ_S, γ_Q and the $O(\varepsilon)$ term of $\theta_d(\beta)$.